Soils and land use

SWAT+ Documentation

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Purpose

`soils.sol` defines every soil used in the watershed. Each soil is a block of lines: one profile line with the soil name, number of layers, hydrologic soil group and other whole-profile values, followed by one line per layer with depth, bulk density, available water capacity, saturated hydraulic conductivity, organic carbon, texture and other layer properties. An HRU picks a soil by name in the `soil` column of `hru-data.hru`. The file is named first on the `soils` line of `file.cio` (default `soils.sol`).

Every HRU should name a soil that is in the file. An unknown name is not reported: the HRU silently gets the first soil in `soils.sol`. Check the `sname` column of `checker.out`, which lists the soil each HRU uses.

The initial nutrient and carbon contents of the soil come from `nutrients.sol`, which an HRU selects through `soil_plant.ini`. The same `soils` line of `file.cio` names `nutrients.sol` second and `soils_lte.sol` third (see soils_lte.sol below).

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each soil then takes `nly + 1` lines: the profile line and `nly` layer lines. The reader counts soils by reading each profile line and skipping the next `nly` lines, so `nly` must equal the number of layer lines that follow. Columns are read by

position; the header names are not checked. Spacing is not significant, so the wide indentation the SWAT+ Editor gives the layer lines has no meaning.

Profile line

#	Column	Units	Used if zero or out of range	Description
1	`name`	-	-	Soil name, up to 20 characters, matched against the `soil` column of `hru-data.hru`.
2	`nly`	-	-	Number of layer lines that follow. There is no upper limit in the model.
3	`hyd_grp`	-	-	Hydrologic soil group: `A`, `B`, `C` or `D`. It selects the curve number from `cntable.lum` and the conductivity used when `soil_k` is zero. Dual groups such as `A/D` and lower-case letters are not recognized.
4	`dp_tot`	mm	-	Maximum rooting depth. See Rooting depth for how 62.0.1 uses it.
5	`anion_excl`	fraction	0.5 if 0; 0.99 if 1 or more	Fraction of porosity from which anions are excluded.
6	`perc_crk`	fraction	-	Potential (maximum) crack volume of the soil profile as a fraction of total soil volume. Used only when crack flow is switched on (`crack` = 1 in `codes.bsn`).
7	`texture`	-	-	Texture label, up to 16 characters. Only the first whitespace-delimited word is read, so `L-L-L-L` is read whole but `L- L- L- L` is read as `L-`. The label is not used in any process; it can serve as a condition in `calibration.cal` (condition `texture`).

Layer lines

Layers are listed from the surface down. Each line has fourteen values.

#	Column	Units	Used if zero or out of range	Description
1	`dp`	mm	-	Depth from the soil surface to the bottom of the layer.
2	`bd`	Mg/m³	1.3 if 0; capped at 2.0	Moist bulk density: oven-dry mass divided by the soil volume at or near field capacity.
3	`awc`	mm H_2O/mm soil	0.005 if 0; capped at 0.8	Available water capacity: water held between field capacity (-0.033 MPa) and permanent wilting point (-1.5 MPa).
4	`soil_k`	mm/h	by `hyd_grp` if 0 or less: A 50, B 20, C 5, D 2, other 10	Saturated hydraulic conductivity.

#	Column	Units	Used if zero or out of range	Description
5	`carbon`	% of soil mass	0.001 if below 0.001	Organic carbon content of the fine earth (material passing a 2 mm sieve).
6	`clay`	%	-	Clay, particles below 0.002 mm.
7	`silt`	%	-	Silt, particles between 0.002 and 0.05 mm.
8	`sand`	%	-	Sand, particles between 0.05 and 2.0 mm.
9	`rock`	% of total mass	capped at 98	Rock fragments, particles above 2 mm.
10	`alb`	fraction	-	Moist soil albedo. Only the top layer value is used.
11	`usle_k`	0.013 t m^2 $h/(m^3$ t cm)	-	USLE soil erodibility factor. Only the top layer value is used.
12	`ec`	dS/m	-	Electrical conductivity. Read and reported, but not used by any process in 62.0.1.
13	`caco3`	%	2.8 if 0	Calcium carbonate content.
14	`ph`	-	6.5 if 0	Soil pH. Read and reported, but not used by any process in 62.0.1.

`clay`, `silt` and `sand` are percentages of the mineral fraction and should add up to 100. The model does not check this, except that in sub-daily runs the sand content of the top layer is reset to 100 − clay − silt.

Layering

Thin first layer

After reading a soil, the reader checks the first layer. If it is shallower than 20 mm and the soil has only one layer, or the second layer reaches deeper than 20 mm, the bottom of the first layer is set to 20 mm.

Added 10 mm surface layer

Without a `soil_lyr_depths.sol` file in the run folder (the usual case), SWAT+ adds a 10 mm surface layer to every soil whose first layer is deeper than 19.5 mm. The new layer takes all its properties from the first input layer, so a soil with four input layers is simulated with five. The first input layer then runs from 10 mm to its listed depth. Outputs by layer count this added layer.

If the first layer is 19.5 mm or shallower (possible only when the second layer is also within 20 mm), no layer is added and the first layer is given a depth of 10 mm.

soil_lyr_depths.sol

If a file named `soil_lyr_depths.sol` is in the run folder, every soil is re-layered to the depths it lists. The file is not named in `file.cio`; SWAT+ looks for this name.

- Line 1 is a title, line 2 a header and line 3 a units label; all three are skipped.
- Each later line gives one depth in mm, from the surface down, as a whole number.
- Depths of 10 mm or less are skipped. The first layer is always 0 to 10 mm.
- A depth beyond the bottom of a soil is replaced by its total depth, and the list ends there for that soil. If the deepest listed depth is shallower than a soil, the profile is cut off at that depth.

Each new layer takes the depth-weighted average of the input layers it overlaps, computed millimetre by millimetre, for every layer property. `refdata/Ames_sub1/soil_lyr_depths.sol` lists 10, 25, 50, 100, 150, 200, 300, 450, 600,

750, 900, 1050, 1200, 1500, 1800 and 2100 mm, so a 2000 mm soil gets sixteen layers with the last from 1800 to 2000 mm.

Septic biozone layer

In an HRU with a septic system (the `sep` column of `landuse.lum`), the profile is split further so that the biozone set in `septic.str` is a separate layer.

Derived layer properties

At the start of the run SWAT+ derives the water contents of each layer from `bd`, `awc` and `clay`. As volume fractions:

$$\theta_{wp} = 0.40\,\frac{c\,\rho_b}{100}, \qquad \theta_{fc} = \theta_{wp} + AWC, \qquad \phi = 1 - \frac{\rho_b}{2.65}$$

where c is `clay` (%), ρ_b is `bd` and AWC is `awc`. If θ_{wp} is zero (no clay) it is set to 0.005. If θ_{fc} reaches the porosity ϕ, field capacity is set to $\phi - 0.05$ and the wilting point to $\theta_{fc} - AWC$; if that is not positive, field capacity becomes $0.75\,\phi$ and the wilting point $0.25\,\phi$.

For a layer of thickness h (mm), the plant-available water at field capacity is $FC = h\,(\theta_{fc} - \theta_{wp})$, the water between wilting point and saturation is $UL = h\,(\phi - \theta_{wp})$, and the percolation travel time (h) is

$$TT_{perc} = \frac{UL - FC}{K_{sat}}$$

with a minimum of 1 hour, where K_{sat} is `soil_k`. The profile sums of FC and UL are reported as `sumfc` and `sumul` in `checker.out`.

The crack volume of each layer is

$$V_{crk} = \text{perc_crk} \times 0.916\,e^{-0.0012\,z}\,h$$

where z is the depth to the bottom of the layer (mm).

The top layer's sand, silt and clay set the particle-size make-up of eroded sediment (sand, silt, clay, small and large aggregates). For the Green and Ampt infiltration method, the wetting front suction head is estimated from the porosity, clay and sand of the top layer.

Use of the soil properties

Initial soil water

Each layer starts at a fraction of its field capacity water. The fraction is `sw_init` in `parameters.bsn`. If `sw_init` is zero, it is estimated from the average annual precipitation P (mm) of the HRU's weather generator:

$$f = \frac{P}{P + \exp(9.043 - 0.002135\,P)}$$

The `sw_frac` column of `soil_plant.ini` is read but not used for this in 62.0.1.

Rooting depth

`dp_tot` limits root depth only while plants that start the run already growing are initialized. After that the model sets the HRU's maximum rooting depth to the bottom of the soil profile. Each planting operation then lowers it to the smaller of its current value and the plant's maximum root depth in `plants.plt`. The value is never raised again during the run, so after a shallow-rooted crop is planted, later crops in the same HRU are limited to that depth as well. The `zmx` column of `checker.out` shows this profile depth. It does not show `dp_tot`.

Runoff, percolation and solute leaching

`hyd_grp` picks one of the four curve numbers in `cntable.lum` for the HRU's land use. `anion_excl` reduces the pore volume available to nitrate, salts and other constituents moving down with percolating water. The nitrate that moves with a water volume w (mm) leaving a layer is

$$NO3_{mobile} = NO3_{ly}\left[1 - \exp\left(\frac{-w}{(1 - \text{anion_excl})\,UL}\right)\right]$$

See hydrology, soil water and temperature and nutrient cycling.

Erosion

The MUSLE soil factor of an HRU uses the top layer only:

$$K \cdot CFRG = \text{usle_k}_1 \; e^{-0.053\,\text{rock}_1}$$

which is multiplied by the support practice factor from `cons_practice.lum` and the slope length and steepness factor. See erosion and sediment.

Albedo

On days with less than 0.5 mm of snow, the albedo of a bare HRU is the top layer `alb`. With plant cover it is $0.23\,(1 - e) + \text{alb}_1\, e$, where $e = \exp(-5 \times 10^{-5}(CV + 0.1))$ and CV is above-ground biomass plus residue (kg/ha). There is no lower limit on `alb`, so a zero value gives zero albedo for bare soil.

Carbon and nutrients

`carbon`, `bd` and the layer thickness set the starting organic matter of each layer: the soil mass is $10\,000\,h\,\rho_b$ kg/ha, organic carbon is `carbon` percent of that, and organic matter is carbon divided by 0.58. How this is split into pools depends on the carbon model (the `carbon` column of `codes.bsn`) and on `nutrients.sol`; see nutrients.sol. In 62.0.1 the 0.001% floor on `carbon` applies to every layer; earlier revisions applied it to the top layer only.

With the Vadas and White phosphorus model (`soil_p` = 1 in `codes.bsn`), `clay` and `carbon` set the initial phosphorus sorption coefficient and `caco3` sets the rate of transfer between the active and stable mineral P pools:

$$k_{as} = 0.0023\,\text{caco3} + 0.005, \qquad 0.002 \le k_{as} \le 0.05$$

`caco3` is not used with the original phosphorus model.

Calibration and conditions

Soil columns can be changed in `calibration.cal` with these parameter names, for a range of layers: `anion_excl`, `crk` (`perc_crk`), `z` (`dp`), `bd`, `awc`, `k` (`soil_k`), `cbn` (`carbon`), `clay`, `silt`, `sand`, `rock`, `alb`, `usle_k`, `ec`, `cal` (`caco3`) and `ph`. Changes to `z`, `bd`, `awc` and `clay` recompute the layer water contents and the curve number coefficients. See parameter change files.

Calibration changes are applied before the soil organic pools are set up, and that setup works from the `carbon` column. A `cbn` change acts on the layer's organic carbon pool, so it is replaced when the pools are initialized and has no lasting effect in 62.0.1.

The `calibration.cal` conditions `hsg` and `texture` select HRUs by `hyd_grp` and `texture`. Decision tables can test `hyd_soil_group` and `k_factor` (the top layer `usle_k`).

soils_lte.sol

`soils_lte.sol` is used only by HRU-LTE objects (`hru-lte.hru`), which select a soil texture by name instead of a layered profile. Line 1 is a title and line 2 a header. The next twelve lines are read, one per texture class, each with four values:

#	Column	Units	Description
1	`texture`	-	Texture class name, matched against the soil texture column of `hru-lte.hru`.
2	`awc`	mm/mm	Available water capacity, multiplied by the soil depth given in `hru-lte.hru`.
3	`por`	mm/mm	Porosity, multiplied by the same soil depth.
4	`scon`	mm/h	Saturated hydraulic conductivity.

Twelve lines are always read. A project with HRU-LTE objects needs this file, since there is no built-in texture table.

Example

`refdata/Osu_1hru/soils.sol` in the SWAT+ repository, first soil, with the spacing reduced:

```
soils.sol: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name     nly hyd_grp     dp_tot anion_excl perc_crk texture  dp      bd     awc   soil_k  carbon   clay   s
SONGSAN    5       A 1200.00000    0.50000  0.50000 LS
                                                              180.0 1.39000 0.10871 16.10093 1.41760 17.300 25
                                                              250.0 1.64000 0.09959 19.31345 0.58820 15.700 25
                                                              460.0 1.48000 0.10069 18.05822 0.42350 16.100 20
                                                              820.0 2.00000 0.07661 37.49110 0.20000 11.300 18
                                                             1200.0 2.00000 0.06645 59.38071 0.26470  7.800 15
```

A loamy sand in hydrologic group A with five layers to 1200 mm. Because the first layer reaches 180 mm, SWAT+ adds a 10 mm surface layer and simulates six layers. `caco3` and `ph` are zero, so the model uses 2.8% and 6.5. The albedo (0.20) and K factor (0.24977) of the first layer are the only values of those two columns that affect the run.

Related files

- `hru-data.hru`: selects a soil by name in its `soil` column.
- `soil_plant.ini` and `nutrients.sol`: initial soil water, nutrients and other constituents of the soil.
- `cntable.lum`: curve numbers chosen by `hyd_grp`.
- `plants.plt`: plant maximum root depth, which limits rooting together with the soil depth.
- `codes.bsn` and `parameters.bsn`: crack flow switch, phosphorus model, carbon model and `sw_init`.
- `septic.str`: septic biozone layer.
- `checker.out`: profile summary by HRU.

Sources

- Readers: `soil_db_read.f90`, `soil_lte_db_read.f90`, called from `proc_read.f90`
- Type definitions: `type soil_profile_db`, `type soilayer_db` and `type soil_lte_database` in `soil_data_module.f90`; simulated soil in `soil_module.f90`
- Name lookup from `hru-data.hru`: `hru_read.f90`
- Layering, `soil_lyr_depths.sol` and septic split: `soils_init.f90`, `layersplit.f90`
- Value checks and derived properties: `soil_phys_init.f90`, `soil_awc_init.f90`, `soil_text_init.f90`
- Initial soil water, rooting depth and MUSLE factor: `hydro_init.f90`, `mgt_plantop.f90`, `pl_root_gro.f90`
- Curve number: `cn2_init.f90`, `curno.f90`
- Albedo, leaching and P: `albedo.f90`, `nut_nlch.f90`, `nut_pminrl2.f90`
- Carbon and nutrient start values: `soil_nutcarb_init.f90`, called from `main.f90.in`

- HRU-LTE texture lookup: hru_lte_read.f90
- Calibration and conditions: cal_parm_select.f90, cal_conditions.f90, conditions.f90
- File name slots: in_sol%soils_sol, in_sol%nut_sol, in_sol%lte_sol in input_file_module.f90

Column	Fortran variable
name	soildb%s%snam
nly	soildb%s%nly
hyd_grp	soildb%s%hydgrp
dp_tot	soildb%s%zmx
anion_excl	soildb%s%anion_excl
perc_crk	soildb%s%crk
texture	soildb%s%texture
dp	soildb%ly%z, then sol%phys%d
bd	soildb%ly%bd, then sol%phys%bd
awc	soildb%ly%awc, then sol%phys%awc
soil_k	soildb%ly%k, then sol%phys%k
carbon	soildb%ly%cbn, then sol%phys%cbn and soil1%cbn
clay, silt, sand, rock	soildb%ly%clay, %silt, %sand, %rock, then sol%phys%...
alb, usle_k, ec	soildb%ly%alb, %usle_k, %ec, then sol%ly%...
caco3	soildb%ly%cal, then sol%ly%cal
ph	soildb%ly%ph, then sol%ly%ph
soils_lte.sol columns	soil_lte%texture, %awc, %por, %scon

Purpose

landuse.lum lists the land use and management records of the project. Each row has a name and a set of names that point to records in other files: the plant community in plant.ini, the management schedule in management.sch, the curve numbers in cntable.lum, the support practice in cons_practice.lum, the urban land type in urban.urb, the Manning's n in ovn_table.lum and the structural practices in the .str files. An HRU picks a row by name in the lu_mgt column of hru-data.hru. The file is named first on the lum line of file.cio (default landuse.lum).

If an HRU names a land use that is not in the file, it silently gets the first row of landuse.lum; nothing is reported.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one land use with fourteen values, read by position. Every column must be filled; write null for an unused link. Names are up to 40 characters.

#	Column	Points to	Description
1	name	-	Land use and management name, matched against lu_mgt in hru-data.hru and named by the decision table action lu_change. Names must be unique.
2	cal_group	-	Calibration group label, called cal_grp in earlier documentation. See Calibration groups.
3	plnt_com	plant.ini	Plant community at the start of the run. null starts the HRU with no plants (bare soil or an urban surface).
4	mgt	management.sch	Management schedule. null means no scheduled operations: no planting, harvest or kill, so only plants that start the run growing (lc_status y in plant.ini) grow.
5	cn2	cntable.lum	Curve number record. The HRU's soil group picks one of its four values.
6	cons_prac	cons_practice.lum	Conservation practice, which sets the USLE P factor.
7	urban	urban.urb	Urban land type. null for land that is not urban.

#	Column	Points to	Description
8	urb_ro	-	Urban build-up and wash-off method: usgs_reg (USGS regression equations) or buildup_washoff (build-up and wash-off, as in SWMM). Any other value, including null, gives no urban sediment and nutrient loads even when urban is set.
9	ov_mann	ovn_table.lum	Overland flow Manning's n class.
10	tile	tiledrain.str	Tile drainage.
11	sep	septic.str	Septic system. See Septic systems.
12	vfs	filterstrip.str	Vegetative filter strip.
13	grww	grassedww.str	Grassed waterway.
14	bmp	bmpuser.str	User-defined BMP removal efficiencies.

Name matching

All names are matched case-sensitively when the files are read, before the HRUs are set up. For each name other than null that is not found, SWAT+ writes a line to diagnostics.out with the land use name and the missing name:

Column	Message	Result when not found
plnt_com	not found in plant.ini	No plants
mgt	not found in management.sch	No schedule
cn2	not found in cntable.lum	Built-in curve numbers 30, 55, 70, 77 (soil groups A to D)
cons_prac	not found (landuse-consprac)	P factor 1.0
tile	not found (landuse-tiledr)	No tile drain
sep	not found (landuse-septic)	No septic system
vfs	not found (landuse-fstrip)	No filter strip
grww	not found (landuse-grassww)	No grassed waterway
bmp	not found (landuse-bmpuser)	No user BMP

urban and ov_mann are matched later, when each HRU's plant community is set up, and an unknown name is not reported. An unknown urban name leaves the HRU non-urban. An unknown ov_mann name leaves the HRU's Manning's n at its built-in value of 0.05.

Use at the start of the run

For each HRU, SWAT+ takes from its land use:

- the plant community, the management schedule and the starting operation in the schedule (from rot_yr_ini in plant.ini);
- the curve number for moisture condition II from cntable.lum, for the soil's hydrologic group, from which the condition I and III values follow (see hydrology);
- the USLE P factor from cons_practice.lum;
- the urban land type and urban runoff method;
- the overland Manning's n (the ovn_mean column of ovn_table.lum), used for the time of concentration;
- the structural practice parameters from the .str files. The septic record is assigned before the soil layers are set up, so that a biozone layer can be added to the profile.

Land use change

The decision table action `lu_change` (see decision tables) switches an HRU to another `landuse.lum` row during the run and logs the change to `lu_change_out.txt`. The new plant community, schedule, curve number, P factor (and the USLE multiplier), urban type and Manning's n replace the old ones. Some values set at the start are not recalculated:

- The time of concentration keeps the old Manning's n.
- The structural practice parameters (tile, filter strip, grassed waterway, user BMP, septic) are not set again. The HRU keeps those of its first land use.
- The `cal_group` used by `calibration.cal` and decision table conditions stays that of the first land use, while soft calibration uses the group of the current land use.

Calibration groups

`cal_group` labels land uses for calibration:

- Soft calibration matches the `name` column of the land use regions in `water_balance.sft` against `cal_group`; the name `basin` matches every HRU.
- In `calibration.cal`, the condition `cal_group` limits a parameter change to HRUs with that group, and the condition `landuse` limits it to HRUs with that `landuse.lum` name.
- Decision tables can test `cal_group` as a condition.

A value of `null` is an ordinary label: all land uses with `null` form one group.

Septic systems

When `sep` names a record in `septic.str`, that record is assigned to every HRU with this land use. If its operation flag is not zero, SWAT+ adds a biozone layer to the soil profile at the depth and thickness given in `septic.str`, and from the year the system becomes operational it runs the biozone routine on each day the biozone layer is above freezing.

In 62.0.1 the daily volume of septic tank effluent is never calculated and stays at zero. No effluent water is added to the biozone layer, and the effluent concentrations in `septic.sep`, which are multiplied by that volume, add no nitrogen, phosphorus or BOD to the soil. The biozone routine still runs on the water and nutrients already in the layer. Septic systems set up through this column therefore have no effluent load on the HRU water or nutrient balance.

Example

`refdata/Ames_sub1/landuse.lum` in the SWAT+ repository, four of its rows, with the spacing reduced:

```
landuse.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name      cal_group plnt_com  mgt      cn2        cons_prac     urban urb_ro          ov_mann        tile
cosy_lum  null      cosy_comm mgt_01   rc_strow_g up_down_slope null  null            convtill_nores null
agrr_lum  null      agrr_comm AGRR_rot rc_strow_g cross_slope   null  null            convtill_res   null
frst_lum  null      frst_comm null     wood_f     up_down_slope null  null            forest_med     null
urld_lum  null      null      null     urban      up_down_slope URLD  buildup_washoff urban_asphalt  null
```

- `cosy_lum` grows the corn and soybean community `cosy_comm` under schedule `mgt_01`, with row crop curve numbers (straight row, good condition) and no support practice (P = 1 for up-and-down-slope tillage).
- `frst_lum` has no schedule, so the forest in `frst_comm` has no operations; it grows only if `frst_comm` starts it growing.
- `urld_lum` has no plants. It links to the urban type `URLD` in `urban.urb`, uses build-up and wash-off for urban loads, the `urban` curve numbers (98 for all soil groups) and the Manning's n of asphalt.
- In this project `AGRR_rot` is not in `management.sch`, so `diagnostics.out` reports `agrr_lum AGRR_rot not found in management.sch` and HRUs with `agrr_lum` run without a schedule.

Related files

- hru-data.hru: selects a land use by name in its lu_mgt column.
- plant.ini: plant communities.
- management.sch: management schedules.
- cntable.lum, [cons_practice.lum](#input-reference-lum-tables-cons-practicelum) and ovn_table.lum: the lookup tables.
- Structural practices and urban.urb: tiledrain.str, septic.str, filterstrip.str, grassedww.str, bmpuser.str, septic.sep and urban.urb.
- lum.dtl: decision tables, including lu_change.
- Plant growth and land use management.

Sources

- Reader: landuse_read.f90, called from proc_db.f90 after the plant community, schedule, lookup table and structural files
- Type definitions: type land_use_management and type land_use_structures in landuse_data_module.f90
- Name lookup from hru-data.hru: hru_read.f90; cal_group copied to the HRU in hrudb_init.f90
- HRU setup: hru_lum_init.f90, plant_init.f90, cn2_init.f90, time_conc_init.f90
- Structural practices: structure_init.f90, structure_set_parms.f90; septic setup in proc_hru.f90 and soils_init.f90; daily septic code in hru_control.f90, swr_percmain.f90, sep_biozone.f90
- Urban loads: hru_urban.f90
- Land use change: lu_change in actions.f90
- Calibration groups: cal_conditions.f90, conditions.f90, calsoft_hyd.f90
- File name slot: in_lum%landuse_lum in input_file_module.f90

Column	Fortran variable
name	lum%name
cal_group	lum%cal_group, copied to hru%cal_group and hru%lum_group_c
plnt_com	lum%plant_cov, index lum_str%plant_cov
mgt	lum%mgt_ops, index lum_str%mgt_ops
cn2	lum%cn_lu, index lum_str%cn_lu
cons_prac	lum%cons_prac, index lum_str%cons_prac
urban	lum%urb_lu, index hru%luse%urb_lu
urb_ro	lum%urb_ro, copied to hru%luse%urb_ro
ov_mann	lum%ovn, value hru%luse%ovn
tile	lum%tiledrain, index lum_str%tiledrain
sep	lum%septic, index lum_str%septic, then iseptic
vfs	lum%fstrip, index lum_str%fstrip
grww	lum%grasww, index lum_str%grasww
bmp	lum%bmpuser, index lum_str%bmpuser

Purpose

Three lookup tables supply values that landuse.lum names by record: cntable.lum (SCS curve numbers), cons_practice.lum (USLE support practice factor) and ovn_table.lum (Manning's n for overland flow). They are named third, fourth and fifth on the lum line of file.cio, after landuse.lum and management.sch. If a file is missing or named null, every land use gets the built-in values given below.

All three files have the same layout. Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one record, read by position. Values after the last numeric column (the descriptions the SWAT+ Editor writes) are not read. Record names are up to 40 characters and are matched case-sensitively.

cntable.lum

Curve numbers for moisture condition II, one value per hydrologic soil group, for each combination of land cover, treatment and hydrologic condition. The `cn2` column of `landuse.lum` names a record.

#	Column	Units	Built-in value	Range	Description
1	`name`	-	-	-	Curve number record name.
2	`cn_a`	-	30	30 to 100	Curve number for hydrologic soil group A.
3	`cn_b`	-	55	30 to 100	Curve number for hydrologic soil group B.
4	`cn_c`	-	70	30 to 100	Curve number for hydrologic soil group C.
5	`cn_d`	-	77	30 to 100	Curve number for hydrologic soil group D.

The SWAT+ Editor also writes `description`, `treat` and `cond_cov` columns, which are not read. The built-in values apply to land uses whose `cn2` is `null` or not found.

Curve number in the model

At the start of the run each HRU takes the value for the hydrologic soil group (`hyd_grp`) of its soil in `soils.sol`. The table values are those of SCS Engineering Division (1986), which apply to a 5% slope; SWAT+ 62.0.1 does not adjust them for the HRU slope. From CN_2 SWAT+ derives the dry and wet condition values

$$CN_1 = CN_2 - \frac{20\,(100 - CN_2)}{(100 - CN_2) + \exp[2.533 - 0.0636\,(100 - CN_2)]}, \qquad CN_1 \geq 0.4\,CN_2$$

$$CN_3 = CN_2\,\exp[0.006729\,(100 - CN_2)]$$

and the retention parameter curve used for daily runoff, as described in hydrology.

CN_2 can change during the run through the management operation `cnup` in `management.sch`, the decision table action `cn_update` (both limited to 35 to 95), a land use change (`lu_change`, which picks the new land use's record) and the calibration parameter `cn2` in `calibration.cal`.

Example

`refdata/Ames_sub1/cntable.lum` in the SWAT+ repository, first records, with the spacing reduced:

```
cntable.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name        cn_a      cn_b      cn_c      cn_d  description  treat              cond_cov
fal_bare    77.00000  86.00000  91.00000  94.00000  Fallow   Bare_soil          ----
fal_res_p   76.00000  85.00000  90.00000  93.00000  Fallow   Crop_residue_cover  Poor
fal_res_g   74.00000  83.00000  88.00000  90.00000  Fallow   Crop_residue_cover  Good
rc_strow_p  72.00000  81.00000  88.00000  91.00000  Row_crops  Straight_row     Poor
```

The names combine land cover, treatment and condition: `rc_strow_p` is row crops, straight row, poor condition. The SWAT+ Editor's table has 52 records, from fallow and row crops to forest, farmsteads and roads.

cons_practice.lum

USLE support practice factors for conservation practices. The `cons_prac` column of `landuse.lum` names a record.

#	Column	Units	Built-in value	Range	Description
1	`name`	-	-	-	Conservation practice name.
2	`usle_p`	-	1.0	0 to 1	USLE support practice factor P.
3	`slp_len_max`	m	1.0	-	Maximum slope length for the practice. Read but not used (see below).

The SWAT+ Editor also writes a `description` column, which is not read. The built-in values apply to land uses whose `cons_prac` is `null` or not found, so a P factor of 1 means no support practice.

P factor in the model

P enters the MUSLE sediment yield through a multiplier fixed at the start of the run:

$$K \cdot C_{frg} \cdot P \cdot LS \cdot 11.8$$

where K and C_{frg} come from the top soil layer (see soils.sol), and the slope length and steepness factor is

$$LS = \left(\frac{L}{22.128}\right)^{m} \left(65.41 \, \sin^2 \alpha + 4.56 \, \sin \alpha + 0.065\right), \qquad m = 0.6 \left[1 - \exp(-35.835 \, S)\right]$$

with L the slope length (`slp_len` in `topography.hyd`, m), S the slope (m/m) and $\alpha = \arctan S$. The model works out the smaller of L and `slp_len_max`, but then computes LS from L, so `slp_len_max` has no effect on LS in 62.0.1. See erosion and sediment.

P can also be set during the run by the decision table actions `p_factor`, `contour`, `stripcrop` and `terrace`, which recompute the multiplier. The calibration parameter `usle_p` in `calibration.cal` changes the HRU's P factor after the multiplier has been computed and does not recompute it, so by itself it does not change sediment yield.

Example

`refdata/Ames_sub1/cons_practice.lum`, first records:

```
cons_practice.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                 usle_p   slp_len_max  description
up_down_slope       1.00000     121.00000  Up_and_down_slope
cross_slope         0.75000     121.00000  Cross_slope_tillage
contour_farming     0.50000     121.00000  Contour_tillage
strip_cros_slope    0.37000     121.00000  Strip_cropping_cross_slope
```

The SWAT+ Editor's table has 38 records: up-and-down and cross-slope tillage, contouring and strip cropping by slope class (1-2% to 21-25%, with pasture or row crops), and terraces by slope class with sod or underflow outlets.

ovn_table.lum

Manning's roughness coefficient for overland flow by tillage, residue and land cover. The `ov_mann` column of `landuse.lum` names a record.

#	Column	Units	Built-in value	Description
1	`name`	-	-	Manning's n class name.
2	`ovn_mean`	s/m1/3	0.5	Mean Manning's n for overland flow. This is the value the model uses.
3	`ovn_min`	s/m1/3	0.5	Minimum Manning's n. Read but not used; a guide for calibration ranges.
4	`ovn_max`	s/m1/3	0.5	Maximum Manning's n. Read but not used; a guide for calibration ranges.

The SWAT+ Editor also writes a `description` column, which is not read. A land use whose `ov_mann` is `null` or not found does not take a value from this file; its HRUs keep the model's starting value of 0.05.

Manning's n in the model

The HRU's Manning's n sets the overland part of its time of concentration (hours), which controls the peak runoff rate:

$$t_{ov} = \frac{0.0556\,(L\,n)^{0.6}}{(S + 0.0001)^{0.3}}$$

with L the slope length (m) and S the slope (m/m). The same n is used as the roughness of the HRU's concentrated flow channel in the channel part of the time of concentration, and in the area-weighted roughness of routing units.

The time of concentration is computed once, before calibration is applied. The calibration parameter `ovn` in `calibration.cal` changes the HRU's Manning's n after that, and a land use change (`lu_change`) also leaves the time of concentration unchanged, so neither alters the peak runoff rate in 62.0.1.

Example

`refdata/Ames_sub1/ovn_table.lum`, first records:

```
ovn_table.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                 ovn_mean       ovn_min       ovn_max  description
fallow_nores          0.01000       0.00800       0.01200  Fallow_no_residue
convtill_nores        0.09000       0.06000       0.12000  Conventional_tillage_no_residue
convtill_res          0.19000       0.16000       0.22000  Conventional_tillage_residue
chisplow_nores        0.09000       0.06000       0.12000  Chisel_plow_no_residue
```

The SWAT+ Editor's table has 20 records, from fallow (0.01) and tilled cropland through range, grasses and forest (up to 0.80 for heavy forest) to urban surfaces (0.011 for asphalt).

Related files

- `landuse.lum`: names records in these tables in its `cn2`, `cons_prac` and `ov_mann` columns.
- `soils.sol`: the hydrologic soil group that picks the curve number, and the K factor and rock content of the MUSLE multiplier.
- `topography.hyd`: slope and slope length.
- `management.sch` and `lum.dtl`: the `cnup` operation and the `cn_update`, `p_factor`, `contour`, `stripcrop` and `terrace` actions.
- Parameter change files: calibration parameters `cn2`, `usle_p` and `ovn`.

Sources

- Readers: `cntbl_read.f90`, `cons_prac_read.f90`, `overland_n_read.f90`, called from `proc_db.f90`
- Type definitions: `type curvenumber_table`, `type conservation_practice_table` and `type overlandflow_n_table` in `landuse_data_module.f90`
- Name lookup: `landuse_read.f90` (`cn2`, `cons_prac`); `plant_init.f90` (`ov_mann`)
- Curve number: `cn2_init.f90`, `curno.f90`; changes in `mgt_sched.f90` (`cnup`) and `actions.f90` (`cn_update`)
- P factor and LS: `plant_init.f90`, `hydro_init.f90`, `ero_ysed.f90`, `actions.f90`
- Manning's n: `time_conc_init.f90`
- Calibration: `cal_parm_select.f90`; call order in `main.f90.in`
- File name slots: `in_lum%cntable_lum`, `in_lum%cons_prac_lum`, `in_lum%ovn_lum` in `input_file_module.f90`

Column	Fortran variable
`cntable.lum name`	`cn%name`
`cn_a`, `cn_b`, `cn_c`, `cn_d`	`cn%cn(1)` to `cn%cn(4)`, copied to `cn2(j)`
`cons_practice.lum name`	`cons_prac%name`
`usle_p`	`cons_prac%pfac`, copied to `hru%lumv%usle_p`
`slp_len_max`	`cons_prac%sl_len_mx`

Column	Fortran variable
`ovn_table.lum name`	`overland_n%name`
`ovn_mean`	`overland_n%ovn`, copied to `hru%luse%ovn`
`ovn_min`, `ovn_max`	`overland_n%ovn_min`, `overland_n%ovn_max`